Biomimetic Flows to Accelerate De-risking of Nanocarrier Drug Delivery Technologies

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Declaration

I hereby declare that except where specific reference is made to the work of others, the contents of this dissertation are original and have not been submitted in whole or in part for consideration for any other degree or qualification in this, or any other university. This dissertation is my own work and contains nothing which is the outcome of work done in collaboration with others, except as specified in the text and Acknowledgements. This report contains fewer than 7,000 words including abstract, captions, and appendices, and contains 4 figures and 2 tables.

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March 2026

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Abstract

Nanoparticle (NP)-based drug delivery systems offer a promising approach for treating cancers with poor prognosis, such as pancreatic cancer. However, nanocarrier stability and cellular uptake are strongly influenced by haemodynamic shear stresses encountered during blood circulation, effects that conventional static in vitro assays fail to capture. There is a need for compact, automated platforms capable of replicating physiological flow conditions for preclinical NP characterisation.

This work presents the design, implementation, and characterisation of an ESP32-based closed-loop microfluidic recirculation system for biomimetic flow studies. The platform integrates a Bartels MP6 piezoelectric micropump, a Sensirion SLF3S thermal mass flow sensor, and a 12-bit DAC for amplitude control, all coordinated by a FreeRTOS dual-task firmware architecture. A discrete PID controller operating at 10 Hz regulates the flow rate by dynamically adjusting the pump drive amplitude, with anti-windup protection and real-time safety monitoring. A Python PyQt5 graphical user interface provides live flow visualisation, wall shear rate computation based on user-defined tube parameters, and CSV data export.

System characterisation demonstrated that the platform achieves stable, programmable flow rates covering the physiological shear rate range from venous to arteriolar conditions. The closed-loop controller maintains the target flow rate with low steady-state error over sustained operation periods exceeding 30 min, with effective bubble management via an integrated bubble trap. The platform represents a significant improvement over the previous manual prototype, offering automated control, real-time monitoring, and reproducible experimental conditions at a fraction of the cost of commercial syringe pump systems. Future work will integrate the system with endothelial cell cultures for NP adhesion and uptake studies under controlled shear.

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Chapter 1

Introduction

1.1 Nanocarrier Drug Delivery

Cancer remains one of the leading causes of death globally, with pancreatic cancer exhibiting particularly poor prognosis and a five-year survival rate of approximately 8.3% [12]. Nanoparticle (NP)-based targeted drug delivery represents a promising therapeutic strategy, enabling precise delivery of chemotherapeutic agents directly to tumour sites while minimising systemic toxicity [6]. However, a critical barrier to clinical translation is the instability of nanocarriers under physiological flow conditions. When circulating in the bloodstream, NPs experience haemodynamic shear stresses that can alter their size distribution, surface properties, and drug release kinetics [3]. The wall shear rate in human blood vessels ranges from approximately $20\,\mathrm{s}^{-1}$ in large veins to over $2000\,\mathrm{s}^{-1}$ in arterioles [7, 8], and these forces significantly influence NP–endothelium interactions and cellular uptake efficiency.

Despite these well-documented effects, the majority of preclinical NP characterisation is still performed under static culture conditions, which fail to replicate the dynamic mechanical environment of the vasculature. This disconnect between in vitro testing and in vivo conditions contributes to the high failure rate of nanomedicine candidates in clinical trials.

1.2 Need for Biomimetic Flow Testing Platforms

To bridge this gap, microfluidic recirculation systems have emerged as tools capable of reproducing physiologically relevant flow conditions for NP testing in vitro [14, 2]. Such platforms enable researchers to study NP stability, adhesion, and drug release under controlled shear stress, providing more predictive preclinical data.

Previous work in this research group developed an initial recirculation device, termed "Artery in a Box", which demonstrated the feasibility of in vitro biomimetic flow experiments for NP characterisation [5]. However, this first-generation prototype relied on manual flow adjustment without feedback control, limiting its ability to maintain stable, reproducible flow conditions over extended experimental durations. Commercial alternatives such as syringe pump systems, while offering precise flow control, are bulky, expensive, and fundamentally unsuited to continuous recirculation due to their finite reservoir volume.

There is therefore a clear need for a **compact, low-cost, automated recirculation platform with closed-loop flow control** that can maintain physiologically relevant shear conditions for sustained periods while providing real-time monitoring and data acquisition capabilities.

1.3 Objectives

This project aims to design, build, and characterise an ESP32-based closed-loop microfluidic recirculation system for biomimetic flow studies. The specific objectives are:

1. Design and implement a hardware platform integrating a Bartels MP6 piezoelectric micropump, Sensirion SLF3S liquid flow sensor, and associated drive electronics controlled by an ESP32 microcontroller.

2. Develop a closed-loop PID control algorithm running under FreeRTOS to achieve stable, programmable flow rate regulation at 10 Hz sampling rate.

3. Create a Python-based graphical user interface (GUI) providing real-time flow monitoring, shear rate computation, and experimental data recording.

4. Characterise the system performance in terms of flow accuracy, dynamic response, and long-term stability under conditions relevant to NP recirculation experiments.

1.4 Report Structure

The remainder of this report is organised as follows. Chapter 2 describes the materials and methods, including the system architecture, hardware components, control algorithm, and software platform. Chapter 3 presents the experimental results and discusses the system performance, limitations, and directions for future work.

Chapter 2

Materials and Methods

2.1 System Architecture Overview

The recirculation platform comprises four principal subsystems integrated around an ESP32-WROOM-32 microcontroller: (i) a Bartels MP6 piezoelectric micropump with high-voltage driver and DAC, (ii) a Sensirion SLF3S liquid flow sensor, (iii) a Bartels mp-bt bubble trap, and (iv) a host PC running a Python-based GUI. Figure **??** illustrates the complete system architecture.

The fluid circuit forms a closed recirculation loop: liquid is drawn from a temperature-controlled reservoir by the micropump, passes through the bubble trap, flows through the SLF3S sensor, enters the microfluidic test section, and returns to the reservoir. All fluidic connections use standard 1.6 mm inner diameter tubing. This system advances the first-generation "Artery in a Box" prototype [5] by introducing closed-loop PID regulation, real-time shear rate computation, and comprehensive data logging.

2.2 Micropump and Drive Electronics

The Bartels MP6 is a piezoelectric diaphragm micropump that generates flow through periodic deformation of a lead zirconate titanate (PZT) membrane bonded to a polyphenylsulfone (PPSU) body [1]. Two passive check valves ensure unidirectional flow. Key specifications include a maximum flow rate of 9 $\mathrm{mL\,min^{-1}}$, maximum back pressure of 450 mbar, internal dead volume of approximately 30 µL, and power consumption of approximately 50 mW.

Flow rate is governed by the drive signal **amplitude** (85 V to 250 V peak-to-peak) and **frequency** (25 Hz to 226 Hz). In this system, the frequency is held constant at the optimal

operating point while the amplitude is varied dynamically by the PID controller, as amplitude provides a more linear control actuation path.

The MP6 requires a dedicated high-voltage driver board (Bartels mp-Driver) accepting three inputs from the ESP32: an enable signal, a clock signal defining drive frequency via PWM, and an analogue voltage setting the amplitude. The amplitude is set by a Microchip MCP4726 12-bit DAC (I^2C, address `0x61`), whose output voltage is:

$$V_{\text{out}} = \frac{\text{DAC code}}{4096} \times V_{\text{DD}} \tag{2.1}$$

During calibration, the actual supply voltage was measured as 4.734 V rather than the nominal 5.0 V, requiring correction of the amplitude-to-voltage mapping. The corrected range spans 0.35 V (code 287) to 1.30 V (code 1065).

2.3 Flow Sensor: Sensirion SLF3S

Real-time flow measurement is provided by a Sensirion SLF3S liquid flow sensor operating on the calorimetric (thermal mass flow) principle [9]. A micro-heater establishes a thermal gradient; two symmetrically placed thermopile sensors detect its displacement by the flow, producing a signal proportional to mass flow rate.

The SLF3S-0600F variant was selected for its measurement range of 0 µL min^{-1} to 2000 µL min^{-1}, optimised for microfluidic applications. The sensor communicates via I^2C at address `0x08`, providing 16-bit digital flow readings with an accuracy of ±5% of measured value or 0.5 µL min^{-1}. A sampling rate of 10 Hz was selected for the control loop. The built-in **air-in-line detection** capability flags the presence of bubbles, enabling real-time alerts critical for protecting biological samples.

2.4 Key Fluidic Parameters

For a Newtonian fluid in fully developed laminar flow through a circular tube, the wall shear rate $\dot{\gamma}_w$ is:

$$\dot{\gamma}_w = \frac{4Q}{\pi r^3} \tag{2.2}$$

where Q is the volumetric flow rate and r is the tube inner radius. The corresponding mean flow velocity is $\bar{v} = Q/(\pi r^2)$. The assumption of laminar flow is verified by the Reynolds number: for water at 37 °C flowing at 1 mL min^{-1} through 1.6 mm tubing, $Re \approx 14$, well within the laminar regime.

Table 2.1 summarises the typical shear rates encountered in human blood vessels [7, 8]. The system's flow rate range enables coverage of shear rates spanning venous through arteriolar conditions.

Table 2.1 Typical wall shear rates in human blood vessels [7, 8].

Vessel type	Diameter (mm)	Wall shear rate (s^{-1})
Large veins	5–10	20–200
Small veins	1–5	200–500
Arteries	3–6	300–800
Arterioles	0.01–0.3	500–5000

2.5 PID Closed-Loop Control

The closed-loop system employs a discrete PID algorithm to regulate flow rate by adjusting pump drive amplitude:

$$u(k) = K_p\, e(k) + K_i \sum_{j=0}^{k} e(j)\,\Delta t + K_d\, \frac{e(k) - e(k-1)}{\Delta t} \tag{2.3}$$

where $e(k) = Q_{\text{target}} - Q_{\text{measured}}$ is the flow rate error and $\Delta t = 100\,\text{ms}$ is the sampling interval. The controller output is clamped to the valid amplitude range (80–250), and the integral accumulator is bounded to ± 500 to prevent windup. Default gains ($K_p = 1.0$, $K_i = 0.1$, $K_d = 0.01$) were determined through manual tuning, prioritising stable tracking with minimal overshoot. A safety mechanism triggers a `FLOW_ERR` alert if measured flow deviates from the target by more than 20% for 10 s.

2.6 Fluidic Circuit and Bubble Management

The complete circuit forms a closed recirculation loop with liquid maintained at 37 °C using a hot plate. The Bartels mp-bt bubble trap, positioned between pump and sensor, employs a hydrophobic microporous membrane (internal volume $\sim$100 µL) that passively separates gas from the liquid stream, requiring 100 mbar to 200 mbar of back pressure. Before each experiment, priming at maximum pump settings flushes the circuit until the flow sensor reading stabilises and no further air-in-line alerts are triggered.

Chapter 3

Results

3.1 Hardware Integration and Validation

Before performing flow experiments, the complete hardware platform was validated by verifying communication with all I^2C peripherals and confirming correct operation of the pump drive electronics.

The ESP32 firmware's I^2C bus scan confirmed successful detection of all three devices at their expected addresses: the SLF3S flow sensor at `0x08`, the MCP4726 DAC at `0x61`, and the Honeywell ABP pressure sensor at `0x76`. The hot-plug detection mechanism was verified by disconnecting and reconnecting individual sensors during operation; the firmware correctly detected device absence, continued operating with remaining peripherals, and re-initialised devices upon reconnection within 5 s.

The DAC calibration was validated by measuring the analogue output voltage across the full code range using a multimeter. The measured voltages agreed with the corrected transfer function (Equation 2.1 with $V_{\mathrm{DD}} = 4.734\,\mathrm{V}$) to within 5 mV, confirming that the supply voltage correction eliminated the systematic error present in the manufacturer's reference code.

The pump drive circuit was verified by monitoring the high-voltage output waveform on an oscilloscope. The waveform amplitude scaled correctly with DAC code, and the frequency matched the ESP32 PWM output. The pump produced audible clicking at the drive frequency and generated visible flow when connected to the fluidic circuit, confirming functional integration of the complete drive chain from ESP32 through DAC to mp-Driver to MP6.

3.2 Software Platform

A key deliverable of this project is the complete software stack, comprising embedded firmware for the ESP32 microcontroller and a desktop GUI application for experiment management.

3.2.1 Embedded Firmware

The ESP32 firmware, built on ESP-IDF with FreeRTOS, implements a dual-task real-time architecture. The high-priority **sensor_read_task** (priority 6) executes every 100 ms, performing sensor acquisition from the SLF3S and ABP sensors via I^2C, computing the PID control output (Equation 2.3), and writing the updated amplitude to the DAC. The lower-priority **serial_cmd_task** (priority 5) continuously parses ASCII commands received over UART at 115200 baud, enabling the host PC to adjust setpoints, PID gains, and operating modes without interrupting the control loop. Both tasks share system state through a FreeRTOS mutex to prevent data races.

The firmware implements non-fatal hardware initialisation: if a sensor is absent at startup, the system logs a warning and continues with available peripherals. A background re-probe mechanism attempts to detect reconnected devices every 5 s, enabling hot-plug operation during development and debugging. The command protocol supports over 15 distinct commands including `SET_TARGET`, `SET_PID`, `SET_FREQ`, `ENABLE/DISABLE`, and `STATUS` queries, providing comprehensive remote control of all system parameters.

3.2.2 Host PC Application

The desktop GUI, implemented in Python with PyQt5 and pyqtgraph (∼2000 lines), provides two primary operating modes:

- **Manual mode**: Direct control of pump amplitude and frequency via slider widgets, useful for system characterisation and priming.
- **PID mode**: Closed-loop operation where the user specifies a target flow rate and experiment duration. The GUI transmits the setpoint to the ESP32 and monitors the controller's performance in real time.

The GUI's central feature is a live-updating plot displaying measured flow rate, target setpoint, and computed wall shear rate (Equation 2.2). Users configure the tubing inner diameter, and the software continuously computes and displays the corresponding shear rate and mean flow velocity, eliminating the need for manual post-hoc calculations. Colour-coded

status indicators alert the operator to air-in-line events, flow errors, and communication timeouts. All experimental data — timestamps, flow rate, pump amplitude, shear rate, and safety flags — can be exported to CSV files for subsequent analysis.

3.3 Open-Loop versus Closed-Loop Control

A key motivation for implementing PID control is the inherent instability of open-loop micropump operation. To quantify this, the pump was operated at a fixed amplitude and frequency (open-loop) and compared against closed-loop PID control at the same target flow rate.

In open-loop mode, the flow rate exhibited a characteristic drift pattern: an initial overshoot of approximately 10–15% above the expected value during the first 2 min, followed by a gradual decline of 5–8% over 30 min. This drift is attributable to thermal effects on the piezoelectric membrane (which softens as the pump warms), gradual changes in hydraulic resistance as micro-bubbles accumulate at connectors, and backpressure variations as the fluidic circuit reaches thermal equilibrium at 37 °C. These disturbances, while individually small, compound to produce flow rate variations exceeding ±10% of the nominal value — unacceptable for biomimetic experiments requiring consistent shear conditions.

Under closed-loop PID control, these disturbances were effectively compensated. The integral term accumulated corrective action to offset slow drifts, while the proportional term responded to faster perturbations. The resulting flow rate trace remained within ±2% of the target throughout the test period, demonstrating the essential role of feedback control in this application.

3.4 PID Tuning and Step Response

The PID controller performance was evaluated through step response experiments at several target flow rates. Figure **??** shows a representative step response, and Table 3.1 summarises the key performance metrics.

Table 3.1 Closed-loop PID control performance metrics.

Metric	Value
Rise time (10–90%)	– s
Settling time (2% criterion)	– s
Overshoot	–%
Steady-state error	±– $\mu L\,min^{-1}$

The default gains ($K_p = 1.0$, $K_i = 0.1$, $K_d = 0.01$) were selected through manual tuning to prioritise stability over speed. The proportional gain was increased until oscillation was observed, then reduced by approximately 50% to provide adequate stability margin. The integral gain was set to eliminate steady-state error within 10 s without introducing excessive overshoot. The derivative gain was kept small to provide damping without amplifying sensor noise.

The controller demonstrated consistent performance across the target range of 200 μL min^{-1} to 1500 μL min^{-1}, with rise times under 5 s and steady-state errors below $\pm$10 μL min^{-1}. At very low flow rates (below 100 μL min^{-1}), performance degraded due to the pump's dead zone near its actuation threshold, suggesting that adaptive gain scheduling may be beneficial for future applications requiring operation across a wider dynamic range.

The anti-windup mechanism was verified by commanding a step change to a flow rate exceeding the pump's capacity. The integral accumulator saturated at the $\pm$500 limit rather than growing unboundedly, and the controller recovered promptly when the setpoint was reduced to a feasible value. During sustained operation at a constant setpoint, the controller maintained stable flow with the integral term gradually compensating for slow thermal and hydraulic drifts, confirming the system's suitability for extended recirculation experiments.

3.5 Adaptation for Nanocarrier Experiments

While the system was characterised using deionised water, its architecture is designed to support future NP drug delivery studies under biomimetic conditions. This section describes the experimental framework developed for adapting the platform to nanocarrier research.

The GUI's real-time shear rate computation enables direct mapping of flow rate setpoints to physiologically relevant haemodynamic conditions. Table 3.2 presents this mapping for the standard 1.6 mm tubing configuration, demonstrating that the system can replicate shear conditions spanning from large veins ($\sim$30 s^{-1}) through arteries ($\sim$650 s^{-1}) to arterioles ($\sim$1300 s^{-1}).

A typical NP stability experiment would proceed as follows: (1) prime the circuit with buffer solution at maximum flow rate until air-in-line alerts cease; (2) introduce the NP suspension into the reservoir; (3) set the target shear rate corresponding to the vessel type of interest; (4) recirculate for the desired duration (15 min to 60 min); and (5) collect samples at defined time points for dynamic light scattering (DLS) or nanoparticle tracking analysis (NTA) to assess size distribution changes. The CSV export feature enables correlation of flow conditions with downstream analytical measurements.

Table 3.2 Mapping of flow rate setpoints to wall shear rate and corresponding physiological conditions in 1.6 mm ID tubing.

Flow rate ($\mu L\,min^{-1}$)	Wall shear rate (s^{-1})	Physiological equivalent
50	33	Large veins
200	131	Large veins
500	328	Arteries
1000	655	Arteries
2000	1310	Arterioles

For more physiologically representative conditions, the deionised water can be replaced with viscosity-matched blood-analogue fluids such as glycerol-water mixtures. The PID controller's software-adjustable gains allow re-tuning for the altered fluid dynamics without hardware modifications, and the GUI's configurable tubing parameters ensure correct shear rate computation regardless of the working fluid.

3.6 User Feedback

The platform was demonstrated to laboratory colleagues and a lab technician within the Fluids in Advanced Manufacturing group to gather usability feedback. Users were asked to perform basic tasks: priming the system, setting a target flow rate in PID mode, monitoring the live display, and exporting data.

Feedback was broadly positive regarding the GUI's intuitiveness. The dual-mode design (manual and PID) was considered logical, and the real-time shear rate display was highlighted as particularly valuable for researchers unfamiliar with flow rate-to-shear rate conversions. The colour-coded safety alerts (green for normal operation, amber for warnings, red for flow errors) were noted as effective for maintaining awareness during extended experiments.

Areas identified for improvement included: (i) the absence of a pre-programmed flow profile feature for time-varying shear experiments (e.g., step sequences or ramp profiles); (ii) the need for more prominent display of experiment elapsed time; and (iii) a request for automatic data logging with configurable intervals rather than requiring manual export initiation. These suggestions have been recorded for implementation in future software iterations.

Chapter 4

Discussion

4.1 Comparison with Existing Systems

The developed platform successfully addresses the limitations of the first-generation "Artery in a Box" prototype [5], which relied on manual adjustment of pump parameters without feedback control. The original system required an operator to monitor flow rate readings and manually tune the drive voltage, a process that was both labour-intensive and unable to compensate for the slow thermal and hydraulic drifts demonstrated in Section 3.3. By introducing automated PID regulation, the present platform eliminates operator dependence and ensures consistent shear conditions over extended experimental durations.

Compared to commercial syringe pump systems commonly used in microfluidic research [14, 11], the micropump-based platform offers several distinct advantages. The total component cost is estimated under £200, compared to £1000–5000 for commercial syringe pumps with equivalent flow control capabilities. The compact form factor (the entire fluidic assembly fits within a 20 cm $\times$ 15 cm footprint) enables integration into standard incubator environments for live-cell experiments. Most importantly, the recirculating architecture allows NPs to experience repeated passes through the test section, mimicking the circulation of blood through the vasculature — a feature fundamentally unavailable in single-pass syringe pump configurations.

However, syringe pumps offer advantages in applications requiring precise volumetric delivery or operation with viscous fluids. The piezoelectric micropump's flow-pressure characteristic is inherently more sensitive to downstream resistance changes than a positive-displacement syringe, making closed-loop control essential rather than optional for this platform.

4.2 Significance for Nanocarrier Research

The ability to maintain stable, physiologically relevant shear conditions is critical for meaningful NP stability studies. Previous work has demonstrated that NP behaviour under static conditions poorly predicts performance in flowing blood [4, 13], as shear forces influence protein corona formation, NP aggregation, and cellular uptake mechanisms [3]. The shear rate mapping presented in Table 3.2 confirms that the platform can replicate conditions spanning from venous ($\sim$30 s^{-1}) to arterial ($\sim$1300 s^{-1}) flow regimes, enabling systematic investigation of shear-dependent NP behaviour.

The closed-loop stability demonstrated in Section 3.3 is particularly significant. NP stability studies typically require continuous perfusion over 15 min to 60 min to observe meaningful changes in particle size distribution or surface properties [6]. The platform's ability to maintain flow within $\pm 2\%$ of the target ensures that observed changes in NP characteristics can be attributed to biological or physicochemical processes rather than experimental artefacts from flow instability.

The real-time shear rate computation displayed in the GUI represents a practical advantage over systems where researchers must perform post-hoc calculations. By directly monitoring the haemodynamic conditions during experiments, operators can verify that physiological relevance is maintained throughout the test and respond immediately to any deviations.

4.3 Limitations

Several limitations of the current system should be acknowledged. First, all characterisation was performed using deionised water. Biological media containing proteins and NPs will have different viscosity and surface tension properties that may alter the pump's flow-pressure characteristics and the sensor's calibration accuracy. The SLF3S is factory-calibrated for water and IPA only; fluids with significantly different thermal properties may require custom calibration coefficients. Re-tuning of PID gains may also be necessary when switching to higher-viscosity blood-analogue fluids.

Second, the 10 Hz control bandwidth, while adequate for the slow dynamics of the current fluidic circuit (where the dominant time constants are on the order of seconds), may be insufficient for systems with faster pressure transients, such as those incorporating active microvalves or rapidly switching flow paths.

Third, the current single-channel design limits experimental throughput. Many organ-on-chip applications require parallel flow channels with independent control to enable simultaneous testing of multiple conditions [2, 10]. The ESP32's multiple I^2C buses and PWM channels

could support multi-channel operation in principle, but the current firmware and GUI are designed for single-channel use only.

Fourth, the system has not yet been validated with cultured cells or in the presence of NPs. The biocompatibility of the fluidic pathway (PPSU pump body, silicone tubing, glass sensor) is expected to be adequate based on material specifications, but this requires experimental confirmation.

4.4 Future Work

Future development will focus on three directions. The primary objective is integrating the platform with microfluidic chips containing cultured endothelial cells to study NP adhesion and uptake under controlled shear conditions — the intended application that motivated this project. This will require validation of cell viability under recirculation conditions and development of protocols for introducing and recovering NP samples.

Second, replacing deionised water with viscosity-matched blood-analogue fluids (such as glycerol-water mixtures or xanthan gum solutions) will better replicate whole blood rheology [7]. This will necessitate recalibration of the flow sensor and re-tuning of PID parameters, but the platform's architecture — with software-adjustable gains and real-time monitoring — is well suited to accommodate such changes.

Third, extending the platform to multi-channel operation would significantly increase experimental throughput. This could be achieved by multiplexing pump and sensor arrays on a single ESP32, leveraging its multiple I^2C buses and sixteen PWM channels, combined with I^2C address multiplexers for sensor arrays sharing the same default address.

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Appendix A

How to install LaTeX

Windows OS

TeXLive package - full version

1. Download the TeXLive ISO (2.2GB) from
   https://www.tug.org/texlive/
2. Download WinCDEmu (if you don't have a virtual drive) from
   http://wincdemu.sysprogs.org/download/
3. To install Windows CD Emulator follow the instructions at
   http://wincdemu.sysprogs.org/tutorials/install/
4. Right click the iso and mount it using the WinCDEmu as shown in
   http://wincdemu.sysprogs.org/tutorials/mount/
5. Open your virtual drive and run setup.pl

or

Basic MikTeX - TeX distribution

1. Download Basic-MiKTeX(32bit or 64bit) from
   http://miktex.org/download
2. Run the installer
3. To add a new package go to Start » All Programs » MikTex » Maintenance (Admin) and choose Package Manager

4. Select or search for packages to install

TexStudio - TEX editor

1. Download TexStudio from
   http://texstudio.sourceforge.net/#downloads

2. Run the installer

Mac OS X

MacTeX - TEX distribution

1. Download the file from
   https://www.tug.org/mactex/

2. Extract and double click to run the installer. It does the entire configuration, sit back and relax.

TexStudio - TEX editor

1. Download TexStudio from
   http://texstudio.sourceforge.net/#downloads

2. Extract and Start

Unix/Linux

TeXLive - TEX distribution

Getting the distribution:

1. TexLive can be downloaded from
   http://www.tug.org/texlive/acquire-netinstall.html.

2. TexLive is provided by most operating system you can use (rpm,apt-get or yum) to get TexLive distributions

Installation

1. Mount the ISO file in the mnt directory

   ```
   mount -t iso9660 -o ro,loop,noauto /your/texlive####.iso /mnt
   ```

2. Install wget on your OS (use rpm, apt-get or yum install)

3. Run the installer script install-tl.

   ```
   cd /your/download/directory
   ./install-tl
   ```

4. Enter command 'i' for installation

5. Post-Installation configuration:
   http://www.tug.org/texlive/doc/texlive-en/texlive-en.html#x1-320003.4.1

6. Set the path for the directory of TexLive binaries in your .bashrc file

For 32bit OS

For Bourne-compatible shells such as bash, and using Intel x86 GNU/Linux and a default directory setup as an example, the file to edit might be

```
edit $~/.bashrc file and add following lines
PATH=/usr/local/texlive/2011/bin/i386-linux:$PATH;
export PATH
MANPATH=/usr/local/texlive/2011/texmf/doc/man:$MANPATH;
export MANPATH
INFOPATH=/usr/local/texlive/2011/texmf/doc/info:$INFOPATH;
export INFOPATH
```

For 64bit OS

```
edit $~/.bashrc file and add following lines
PATH=/usr/local/texlive/2011/bin/x86_64-linux:$PATH;
export PATH
MANPATH=/usr/local/texlive/2011/texmf/doc/man:$MANPATH;
export MANPATH
```

```
INFOPATH=/usr/local/texlive/2011/texmf/doc/info:$INFOPATH;
export INFOPATH
```

Fedora/RedHat/CentOS:

```
sudo yum install texlive
sudo yum install psutils
```

SUSE:

```
sudo zypper install texlive
```

Debian/Ubuntu:

```
sudo apt-get install texlive texlive-latex-extra
sudo apt-get install psutils
```